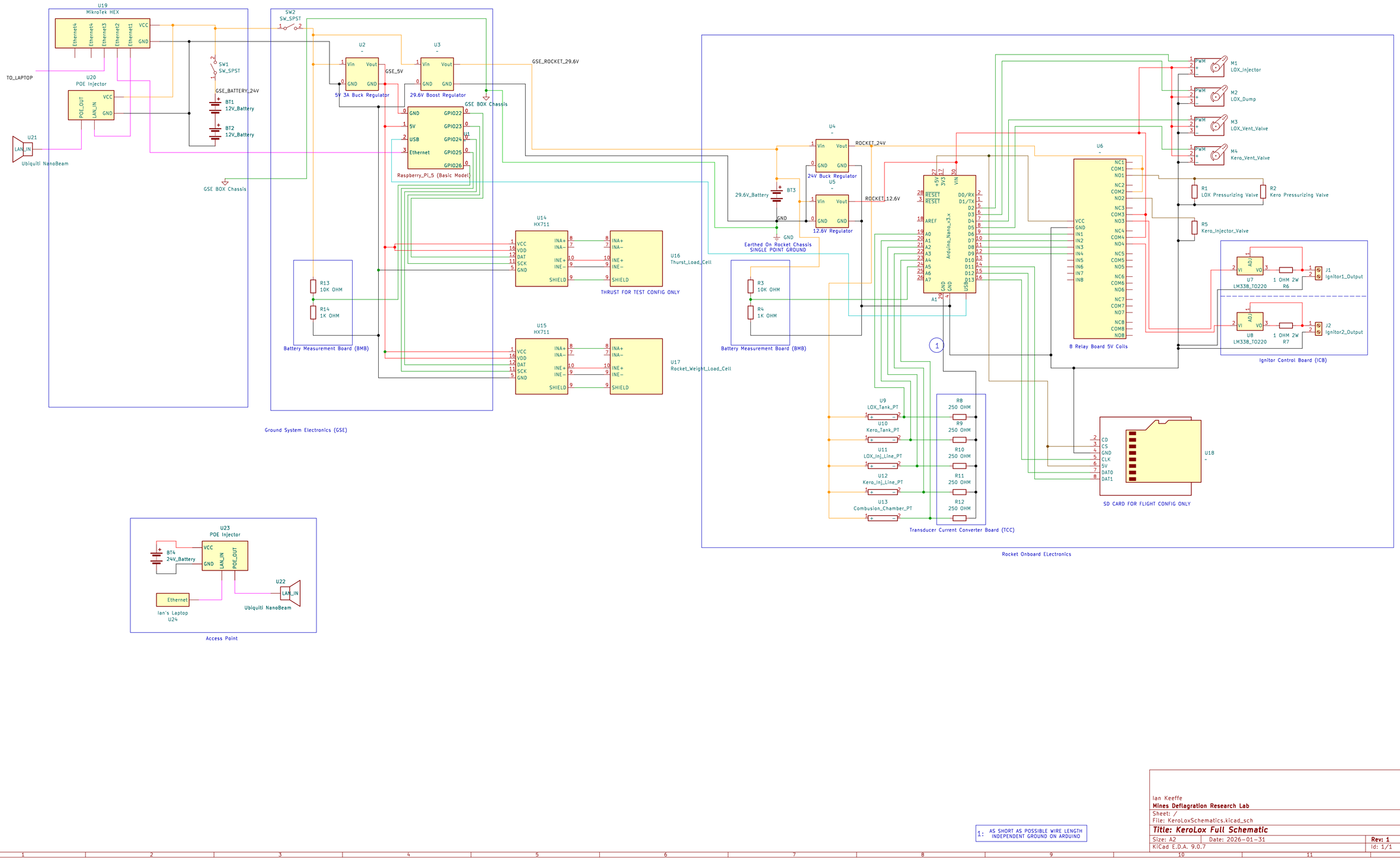

U19
MikroTek HEX
Ethernet4
Ethernet4
Ethernet3
Ethernet2
Ethernet1
VCC
GND
TO_LAPTOP
U20
POE Injector
POE_OUT
LAN_IN
VCC
GND
U21
LAN_IN
Ubiquiti NanoBeam
SW1
SW_SPST
GSE_BATTERY_24V
BT1
12V_Battery
BT2
12V_Battery
GSE BOX Chassis
SW2
SW_SPST
U2
5V 3A Buck Regulator
GSE_5V
U3
29.6V Boost Regulator
GSE_ROCKET_29.6V
GSE BOX Chassis
GND
GPIO22
5V
GPIO23
USB
GPIO24
Ethernet
GPIO25
GPIO26
Raspberry_Pi_5 (Basic Model)
U1
R13
10K OHM
R14
1K OHM
Battery Measurement Board (BMB)
U14
HX711
VCC
VDD
DAT
SCK
GND
INA+
INA−
INE+
INE−
SHIELD
U16
Thurst_Load_Cell
THRUST FOR TEST CONFIG ONLY
U15
HX711
U17
Rocket_Weight_Load_Cell
Ground System Electronics (GSE)
U4
24V Buck Regulator
ROCKET_24V
U5
12.6V Regulator
ROCKET_12.6V
29.6V_Battery
BT3
GND
Earthed On Rocket Chassis
SINGLE POINT GROUND
R3
10K OHM
R4
1K OHM
Battery Measurement Board (BMB)
U6
8 Relay Board 5V Coils
NC1
COM1
NO1
NC2
COM2
NO2
NC3
COM3
NO3
NC4
COM4
NO4
NC5
COM5
NO5
NC6
COM6
NO6
NC7
COM7
NO7
NC8
COM8
NO8
VCC
GND
IN1
IN2
IN3
IN4
IN5
IN6
IN7
IN8
Arduino_Nano_v3.x
RESET
AREF
A0
A1
A2
A3
A4
A5
A6
A7
D0/RX
D1/TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
+5V
3V3
VIN
USB
M1
LOX_Injector
M2
LOX_Dump
M3
LOX_Vent_Valve
M4
Kero_Vent_Valve
R1
LOX Pressurizing Valve
R2
Kero Pressurizing Valve
R5
Kero_Injector_Valve
U7
LM338_TO220
1 OHM 2W
R6
J1
Ignitor1_Output
U8
LM338_TO220
1 OHM 2W
R7
J2
Ignitor2_Output
Ignitor Control Board (ICB)
U9
LOX_Tank_PT
U10
Kero_Tank_PT
U11
LOX_Inj_Line_PT
U12
Kero_Inj_Line_PT
U13
Combusion_Chamber_PT
R8
250 OHM
R9
250 OHM
R10
250 OHM
R11
250 OHM
R12
250 OHM
Transducer Current Converter Board (TCC)
U18
CD
CS
GND
CLK
5V
DAT0
DAT1
SD CARD FOR FLIGHT CONFIG ONLY
Rocket Onboard ELectronics
U23
POE Injector
BT4
24V_Battery
VCC
GND
LAN_IN
POE_OUT
U22
LAN_IN
Ubiquiti NanoBeam
Ethernet
Ian's Laptop
U24
Access Point
1: AS SHORT AS POSSIBLE WIRE LENGTH INDEPENDENT GROUND ON ARDUINO
Ian Keeffe
Mines Deflagration Research Lab
Sheet: /
File: KeroLoxSchematics.kicad_sch
Title: KeroLox Full Schematic
Size: A2
Date: 2026-01-31
Rev: 1
KiCad E.D.A. 9.0.7
Id: 1/1